

Question	Marking Guidance	Mark	Comments
4.1	1. (Most likely to be) transferred to a special care unit are those under 2800 g OR (Most likely to be) transferred to a special care unit are those over 4200 g; 2. Extreme mass babies least likely to survive (to reproduce) and so less likely to pass on their <u>alleles</u> (for extreme mass at birth); 3. Extreme mass at birth decreases in frequency (in the population) OR <u>Alleles</u> (for extreme mass at birth) decrease in frequency (in the population); If neither 1 or 2 awarded allow correct stated mass less/more likely to survive for 1 mark	3	Accept converse answers linked to those with mass at birth at any value between 2800 and 4200 g. 1. For '2800 g' accept any value between 1400 g and 2800 g. 1. For '4200 g' accept any value between 4200 g and 5200 g. 1. If values for both extremes are given, both must be correct. 1. Reject data quoted below 1400 g or above 5200 g. 3. Accept 'proportion/percentage' for 'frequency'. 3. Do not accept 'number' for 'frequency'.
4.2	1. Allele 2. Locus/loci 3. Transcribed 4. Translated 5. Golgi (apparatus)/<u>Rough</u> endoplasmic reticulum 6. Tertiary;;;	3 max	6 correct = 3 marks 4 – 5 correct = 2 marks 2 – 3 correct = 1 mark 0 – 1 correct = 0 marks 2. Do not accept locust. 3. Accept transcribed. 3. Ignore spliced. 5. Reject smooth endoplasmic reticulum. 5. Ignore RER/ER. 6. Ignore 3D. 6. Accept secondary.
4.3	Automarked q – ☒ Chi-squared	1	

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4.4	1. <u>Probability</u> that <u>difference</u> (in frequency of births above 4500 g) is due to <u>chance</u> is less than 0.05 OR <u>Probability</u> that <u>difference</u> (in frequency of births above 4500 g) is due to <u>chance</u> is 0.03; 2. Reject null hypothesis; 3. Presence of <i>KIR2DS1</i>/allele does (significantly) affect the frequency of high birth mass;	3	Ignore reference to critical value. 1. Accept 5% for 0.05 1. Accept 3% for 0.03 1. Ignore results due to chance. 1. Accept '<u>Probability</u> that <u>difference</u> (in frequency of births above 4500 g) is not due to <u>chance</u> is greater than 0.95' OR '<u>Probability</u> that <u>difference</u> (in frequency of births above 4500 g) is not due to <u>chance</u> is 0.97' 2. Accept 'H₀' for null hypothesis. 2. For 'reject' accept 'do not accept' but not 'disprove/wrong'. 2. Accept 'Accept the alternate hypothesis/H₁'. 3. Do not accept 'number' for 'frequency'.

